

Week 3 Assignment

JavaScript Fundamentals (Part 1)

This PDF contains the complete JavaScript code for Tasks 1–6. The programs are organized by task and can be copied into the corresponding .js files.

HTML File: index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Week 3 - JavaScript Fundamentals</title>
</head>
<body>
  <h1>Week 3: JavaScript Fundamentals (Part 1)</h1>
  <p>Open the browser console (F12 → Console) to view the program outputs.</p>

  <script src="task1_basics.js"></script>
  <script src="task2_variables_datatypes.js"></script>
  <script src="task3_operators.js"></script>
  <script src="task4_conditionals.js"></script>
  <script src="task5_loops.js"></script>
  <script src="task6_practice.js"></script>
</body>
</html>
```

Task 1: JavaScript Basics

File: task1_basics.js

```
// Task 1: JavaScript Basics

// Single-line comment
console.log("Task 1: JavaScript Basics");

// Basic JavaScript statements
let studentName = "Student";
let course = "JavaScript Fundamentals";

console.log("Student Name:", studentName);
console.log("Course:", course);

/*
  Multi-line comment:
  JavaScript statements are executed
  by the JavaScript engine.
*/

console.log("JavaScript is linked successfully with the HTML file.");
console.log("This is a basic JavaScript statement.");
```

Task 2: Variables & Data Types

File: task2_variables_datatypes.js

```
// Task 2: Variables & Data Types

console.log("\nTask 2: Variables & Data Types");

// let - value can be changed
let age = 21;
age = 22;

// const - value cannot be reassigned
const country = "India";

// Different data types
let studentName = "Rahul";      // String
let marks = 85;                 // Number
let passed = true;              // Boolean
let futureValue;                // Undefined
let emptyValue = null;          // Null

console.log("Name:", studentName);
console.log("Age:", age);
console.log("Country:", country);
console.log("Marks:", marks);
console.log("Passed:", passed);
console.log("Future Value:", futureValue);
console.log("Empty Value:", emptyValue);

// Checking data types using typeof
console.log("Type of studentName:", typeof studentName);
console.log("Type of marks:", typeof marks);
console.log("Type of passed:", typeof passed);
console.log("Type of futureValue:", typeof futureValue);
console.log("Type of emptyValue:", typeof emptyValue);
```

Task 3: Operators

File: task3_operators.js

```
// Task 3: Operators

console.log("\nTask 3: Operators");

let a = 20;
let b = 5;

// Arithmetic Operators
console.log("Addition:", a + b);
console.log("Subtraction:", a - b);
console.log("Multiplication:", a * b);
console.log("Division:", a / b);
console.log("Modulus:", a % b);
console.log("Exponent:", a ** b);

// Assignment Operators
let number = 10;
number += 5;
console.log("After += 5:", number);
number -= 3;
console.log("After -= 3:", number);
number *= 2;
console.log("After *= 2:", number);
number /= 4;
console.log("After /= 4:", number);

// Comparison Operators
console.log("a == b:", a == b);
console.log("a === b:", a === b);
console.log("a != b:", a != b);
console.log("a > b:", a > b);
console.log("a < b:", a < b);
console.log("a >= b:", a >= b);
console.log("a <= b:", a <= b);

// Logical Operators
let hasId = true;
let hasTicket = true;
console.log("AND:", hasId && hasTicket);
console.log("OR:", hasId || false);
console.log("NOT:", !hasId);

// Increment and Decrement
let count = 5;
count++;
console.log("After increment:", count);
count--;
console.log("After decrement:", count);
```

Task 4: Conditional Statements

File: task4_conditionals.js

```
// Task 4: Conditional Statements

console.log("\nTask 4: Conditional Statements");

// 1. Positive, Negative, or Zero
let number = -8;

if (number > 0) {
    console.log(number, "is positive.");
} else if (number < 0) {
    console.log(number, "is negative.");
} else {
    console.log("The number is zero.");
}

// 2. Even or Odd
let value = 14;

if (value % 2 === 0) {
    console.log(value, "is even.");
} else {
    console.log(value, "is odd.");
}

// 3. Age Eligibility
let age = 20;

if (age >= 18) {
    console.log("Eligible to vote.");
} else {
    console.log("Not eligible to vote.");
}

// 4. Greater of Three Numbers
let first = 45;
let second = 72;
let third = 60;

if (first >= second && first >= third) {
    console.log("Greatest number:", first);
} else if (second >= first && second >= third) {
    console.log("Greatest number:", second);
} else {
    console.log("Greatest number:", third);
}

// 5. Grade Assignment
let marks = 86;

if (marks >= 90) {
    console.log("Grade: A+");
} else if (marks >= 80) {
    console.log("Grade: A");
} else if (marks >= 70) {
    console.log("Grade: B");
} else if (marks >= 60) {
    console.log("Grade: C");
} else if (marks >= 50) {
    console.log("Grade: D");
} else {
    console.log("Grade: F");
}
```

```
// 6. Nested if example
let studentAge = 20;
let hasIdCard = true;

if (studentAge >= 18) {
  if (hasIdCard) {
    console.log("Student is allowed to enter.");
  } else {
    console.log("ID card is required.");
  }
} else {
  console.log("Student is under 18.");
}
```

Task 5: Loops

File: task5_loops.js

```
// Task 5: Loops

console.log("\nTask 5: Loops");

// 1. for loop - numbers from 1 to 10
console.log("Numbers 1 to 10:");
for (let i = 1; i <= 10; i++) {
    console.log(i);
}

// 2. for loop - even numbers
console.log("Even numbers from 1 to 20:");
for (let i = 2; i <= 20; i += 2) {
    console.log(i);
}

// 3. for loop - odd numbers
console.log("Odd numbers from 1 to 20:");
for (let i = 1; i <= 20; i += 2) {
    console.log(i);
}

// 4. Multiplication table
let tableNumber = 5;
console.log("Multiplication table of", tableNumber);

for (let i = 1; i <= 10; i++) {
    console.log(tableNumber + " x " + i + " = " + (tableNumber * i));
}

// 5. while loop - sum of numbers from 1 to 10
let counter = 1;
let sum = 0;

while (counter <= 10) {
    sum += counter;
    counter++;
}

console.log("Sum of numbers 1 to 10:", sum);

// 6. do...while loop - numbers in reverse
let reverseNumber = 10;

console.log("Numbers from 10 to 1:");
do {
    console.log(reverseNumber);
    reverseNumber--;
} while (reverseNumber >= 1);

// 7. Loop with condition
console.log("Numbers divisible by 3 from 1 to 30:");

for (let i = 1; i <= 30; i++) {
    if (i % 3 === 0) {
        console.log(i);
    }
}
```

Task 6: JavaScript Practice Programs

File: task6_practice.js

```
// Task 6: JavaScript Practice Programs
// At least 5 programs combining Week 3 concepts.

// Program 1: Find the sum of even numbers from 1 to 20
let evenSum = 0;

for (let i = 1; i <= 20; i++) {
    if (i % 2 === 0) {
        evenSum += i;
    }
}

console.log("\nPractice 1 - Sum of even numbers:", evenSum);

// Program 2: Check whether a number is prime
let primeNumber = 17;
let isPrime = true;

if (primeNumber < 2) {
    isPrime = false;
} else {
    for (let i = 2; i < primeNumber; i++) {
        if (primeNumber % i === 0) {
            isPrime = false;
            break;
        }
    }
}

console.log("Practice 2 -", primeNumber, "is prime:", isPrime);

// Program 3: Find the factorial of a number
let factorialNumber = 5;
let factorial = 1;

for (let i = 1; i <= factorialNumber; i++) {
    factorial *= i;
}

console.log("Practice 3 - Factorial of", factorialNumber, ":", factorial);

// Program 4: Find the largest number in a set of values
let numbers = [12, 45, 7, 89, 34];
let largest = numbers[0];

for (let i = 1; i < numbers.length; i++) {
    if (numbers[i] > largest) {
        largest = numbers[i];
    }
}

console.log("Practice 4 - Largest number:", largest);

// Program 5: Calculate student result
let studentMarks = 76;
let result;

if (studentMarks >= 50) {
    result = "Pass";
} else {
    result = "Fail";
}
```

```
console.log("Practice 5 - Result:", result);

// Program 6: Print a number pattern using loops
console.log("Practice 6 - Number pattern:");

for (let i = 1; i <= 5; i++) {
  let row = "";

  for (let j = 1; j <= i; j++) {
    row += j + " ";
  }

  console.log(row);
}
```